

Figure 5.10 An example of an AAS record for a Siemens pressure gauge, available from the IDTA AAS server at https://v3.admin-shell-io.com/. Licensed under the Apache License, Version 2.0 http://www.apache.org/licenses/LICENSE-2.0.

Because this pressure gauge carries its own "digital passport". the digital twin can automatically ingest its calibration data without a human typing it in.

5.6 Serving the model of reality

The digital twin’s value lies in its ability to provide a usable, integrated model of reality to clients. This model, built from disparate data and unified by a knowledge graph, must be made accessible and interactive. Our challenge is to present this complex system, which combines data from various stores with a rich relationship network, in a way that is intuitive for users while accurately reflecting the real-world system it represents.

5.6.1 Context-aware data retrieval

To achieve this, we must move beyond traditional data retrieval methods. Standard database queries are limited, requiring explicit identifiers like a specific sensor or room ID. This approach falls short when users need to ask questions that are inherently relational and context-dependent, such as: "What is the temperature in all rooms on the second floor?" or "show me the energy consumption of all HVAC equipment serving the conference rooms".